Sterilizer



Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 8 labs

Covers these lab items: Sterilizer; Sterilizers; Syringe Sterilizer

What it is used for

A sterilizer uses a validated process to destroy or remove microorganisms from suitable instruments and materials. Correct loading, cycle selection, and verification are essential for safe reuse in healthcare and training settings.

Main parts

- Chamber holds the instruments or materials being processed.
- Door and locking mechanism keep the chamber closed during operation.
- Control panel selects the cycle and displays operating status.
- Heating element or steam generator provides the required treatment conditions.
- Temperature and pressure sensors monitor conditions in applicable models.
- Tray or rack holds items for effective exposure to the sterilizing process.
- Drain, vent, or filter system manages moisture or exhaust according to the design.

Before you start

1. Read the equipment instructions and confirm it is designed for the items being processed.
2. Only trained and authorised personnel should operate the sterilizer.
3. Clean instruments first according to the approved infection-control procedure.
4. Inspect the chamber, door seal, trays, controls, and power or utility connections.
5. Use compatible packaging and arrange items to allow the sterilizing agent to reach surfaces.
6. Check water, indicator supplies, and other requirements specified for the selected cycle.

How to use

1. Load cleaned items according to the manufacturer's instructions without overpacking the chamber.
2. Separate incompatible materials and ensure packages do not obstruct vents or sensors.
3. Close and lock the door using the specified procedure.
4. Select the validated cycle suitable for the load and instrument material.
5. Start the cycle and monitor displayed temperature, pressure, time, and alarms as applicable.
6. Wait until the full cycle finishes and the equipment indicates safe completion.
7. Allow pressure to return to a safe level and the load to cool as instructed.
8. Use appropriate protective equipment when unloading hot items or handling residual moisture.
9. Check cycle records and required chemical, physical, or biological indicators according to local procedures.
10. Store processed items in the designated clean area and report any failed cycle.

⚠ Safety

- Never open the door while the chamber is pressurised or the cycle is active.
- Beware of hot surfaces, steam, hot liquids, and sharp instruments during unloading.
- Do not process sealed containers or incompatible materials unless the equipment specifically permits them.
- Never reuse instruments after an incomplete or failed cycle until they are reprocessed correctly.
- Do not bypass door locks, alarms, sensors, or other safety interlocks.
- Follow the facility's infection-control procedures for contaminated items and failed loads.

Care & storage

- Switch off and isolate the equipment before routine cleaning when required by the manual.
- Clean the chamber, trays, and seals using approved methods and compatible products.
- Inspect seals, filters, drains, and indicators at the required intervals.
- Arrange authorised servicing and calibration according to the manufacturer's schedule.

Common mistakes to avoid

- Overloading the chamber or blocking the sterilizing agent's circulation.
- Selecting a cycle unsuitable for the instrument or material.
- Opening the door before pressure has returned to a safe level.
- Assuming an item is sterile without checking cycle completion and required indicators.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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